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$$\lambda = \lim_{x \to \alpha} \frac{|x_{k+1} - x_*|}{|x_k - x_*|}$$

$$= \left| \lim_{x_k \to x_*} \frac{g(x_k) - g(x_*)}{x_k - x_*} \right|$$

$$= \left| \frac{dg}{dx} \right|_{x = x_*}$$

$$\begin{cases} = 0 & \text{super} \\ < 1 & \text{conv} \\ \geq 1 & \text{div} \end{cases}$$

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prove that, the convergent rate of new to n's iteration formula is

$$g(x) = x - \frac{f(x)}{f'(x)}$$

$$\alpha = \left| \frac{d}{dx} \{ g(x) \} \right|_{x = x_\alpha}$$

$$= \left| \frac{d}{dx} \left| x - \frac{f(x)}{f'(x)} \right| \right|_{x = x_\alpha}$$

$$= \left| 1 - \frac{f'(x_\alpha) f(x_\alpha) - f(x_\alpha) f''(x_\alpha)}{[f'(x_\alpha)]^2} \right| \quad (f(x_\alpha) = 0)$$

$$= \left| 1 - \frac{[f'(x_\alpha)]^2}{[f'(x_\alpha)]^2} \right|$$

$$= |1 - 1| = 0$$

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## total num. of operation in triangle

1 + (n-1) + (n-1) = 1 + 2(n-1)

$$\text{operation} = \sum_{i=1}^{n} [1 + 2(i-1)]$$

$$= \sum_{i=1}^{n} [2i - 1)$$

$$= 2 \times \frac{n(n+1)}{2} = n$$

$$= n^2 + n - n$$

$$= n^2$$

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num of operation in Gaussian

num add

$$\frac{n(n-1)(2n-1)}{6}$$

num of multi

$$\frac{n(n-1)(2n-1)}{6}$$

num div

$$\frac{n(n-1)}{2}$$

$$\text{num of opp} = \frac{n(n-1)(2n-1)}{6} + \frac{n(n-1)(2n-1)}{6}$$

$$= \frac{n(n-1)(2n-1)}{3} + \frac{n(n-1)}{2}$$

$$= \frac{n(n-1)(2n-1)}{3} + \frac{n(n-1)}{2}$$

$$= n(n-1)\left[\frac{(2n-1)}{3} + \frac{1}{2}\right]$$

$$= n(n-1)\left[\frac{4n-2+3}{6}\right]$$

$$= n(n-1)\left[\frac{4n+1}{6}\right]$$

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$$= \frac{n(n-1)(4n+1)}{6}$$

$$= \frac{(n^2-n)(4n+1)}{6}$$

$$= \frac{4n^3+n^2-4n^2-n}{6}$$

$$= \frac{4n^3-3n^2-n}{6}$$

$$= \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n$$

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## QR

### general case of $P_k$ and $u_k$

$$P_k = u_k - \sum_{j=1}^{k-1} (u_k^T q_j) q_j$$

$$q_k = \frac{P_k}{|P_k|}$$

$$P_k = u_k - \sum_{j=1}^{k-1} (u_k^T q_i) q_i$$

$$\Rightarrow |P_k| q_k = u_k - \sum_{j=1}^{k-1} (u_k^T q_i) q_i$$

$$\Rightarrow u_k = |P_k| q_k + \sum_{i=1}^{k-1} (u_k^T q_i) q_i \quad \text{— (3)}$$

$$\Rightarrow q_k^T u_k = |P_k| \underbrace{q_k^T q_k}_{1} + \sum_{i=1}^{k-1} (u_k^T q_i) \underbrace{q_k^T q_i}_{0}$$

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$\Rightarrow q_k^T u_k = |p_k| \text{ or } |p_k| = u_k^T q_k$

$① \Rightarrow |p_k| = u_k^T q_k$

$$u_k = (u_k^T q_k) q_k + \sum_{i=1}^{k-1} (u_k^T q_i) q_i$$

$$= \sum_{i=0}^{k} (u_k^T q_i) q_i$$

Theorem: Any $(m \times n)$ matrix $A$, with $m \geq n$, can be written in the form $A = QR$ where $Q$ is $m \times n$ matrix with orthonormal columns and $R$ is an upper-triangular $n \times n$ matrix.

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$A_{m \times n} = Q_{m \times n} \times R_{n \times n}$

$Q^T A = Q^T Q R$

$\Rightarrow Q^T A = R$

Now,

$Ax = b$

$\Rightarrow A^T A x = A^T b$

$\Rightarrow (QR)^T A x = (QR)^T b$

$\Rightarrow R^T Q^T \cdot QR x = R^T Q^T b$

$\Rightarrow R^T I R x = R^T Q^T b$

$\Rightarrow Rx = Q^T b$

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R matrix is upper triangular matrix [ derivation ]

$$q_k = \frac{p_k}{|p_k|}$$

$$p_k = |p_k| \cdot q_k$$

$$p_1 = |p_1| q_1 = u_1$$

$$p_2 = |p_2| q_2 = u_2 - (u_2^T q_1) q_1$$

$$p_3 = |p_3| q_3 = u_3 - (u_3^T q_2) q_2 - (u_3^T q_1) q_1$$

$$p_k = |p_k| q_k = u_k - \sum_{j=1}^{k-1} (u_k^T q_j) q_j$$

now,

$$u_k = |p_k| q_k + \sum_{j=1}^{k-1} (u_k^T q_j) q_j \quad \text{— (1)}$$

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Now,

$$u_k q_k^T = |P_k| \cdot q_k q_k^T + \sum_{i=1}^{k-1} (u_k q_i) q_i + q_k^T$$

$q_k q_k^T$ → 1 orthonon

$(u_k q_i) q_i$ → 0 orthogo

$$u_k \cdot q_k^T = |P_k| \quad \text{——} (11)$$

(1) and (11)

$$u_k = (u_k \cdot q_k^T) q_k + \sum_{i=1}^{k-1} (u_k^T q_i) q_i$$

$$= \sum_{i=1}^{k} (u_k^T q_i) q_i$$

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Now ,

$$u_1 = (u_1^T q_1)\, q_1$$

$$= q_1 (u_1^T q_1)$$

$$= (q_1 \; q_2) \begin{pmatrix} u_1^T q_1 \\ 0 \end{pmatrix}$$

$$u_2 = (u_2^T q_1) q_1 + (u_2^T q_2) q_2$$

$$= (q_1 \; q_2) \begin{bmatrix} u_2^T q_1 \\ u_2^T q_2 \end{bmatrix}$$

$$\begin{bmatrix} u_1 & u_2 \end{bmatrix} = \begin{bmatrix} q_1 & q_2 \end{bmatrix} \begin{bmatrix} u_1^T q_1 & u_2^T q_1 \\ 0 & u_2^T q_2 \end{bmatrix}$$

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Now,

$$u_3 = (u_3^T q_1) q_1 + (u_3^T q_2) q_2 + (u_3^T q_3) q_3$$

$$= [q_1 \; q_2 \; q_3] \begin{bmatrix} u_3^T q_1 \\ u_3^T q_2 \\ u_3^T q_3 \end{bmatrix}$$

$$[u_1 \; u_2 \; u_3] = [q_1 \; q_2 \; q_3] \begin{bmatrix} u_1^T q_1 & u_2^T q_1 & u_3^T q_1 \\ 0 & u_2^T q_2 & u_3^T q_2 \\ 0 & 0 & u_3^T q_3 \end{bmatrix}$$

So nth term

$$[u_1 \; u_2 \; \cdots \; u_n] = [q_1 \; \cdots \; q_n] \begin{bmatrix} u_1^T q_1 & u_2^T q_1 & \cdots & u_n^T q_1 \\ 0 & u_2^T q_2 & \cdots & u_n^T q_2 \\ 0 & 0 & u_3^T q_3 & u_n^T q_3 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & u_n^T q_n \end{bmatrix}$$

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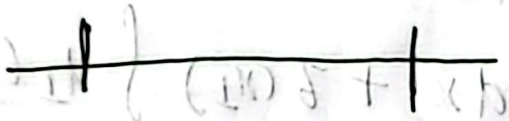
# Trapezium / Trapezoidal

- closed Newton cotes formula with

$n = 1$

number of Nodes $= n + 1$

$= 2$




$$h = \frac{b-a}{n} = \frac{b-a}{1} = b-a$$

$$I_n(f) = \int_a^b P_n(x)\, dx$$

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$$I_1(f) = \int_a^b P_1(x)\,dx$$

$$= \int_a^b \left[ l_0(x) f(x_0) + l_1(x) f(x_1) \right] dx$$

$$= \int_a^b l_0(x) f(x_0)\,dx + \int_a^b l_1(x) f(x_1)\,dx$$

$$= f(x_0) \int_a^b l_0(x)\,dx + f(x_1) \int_a^b l_1(x)\,dx$$

$G_0$ $G_1$

$$G_0 = \int_a^b l_0(x)\,dx \qquad x_0 = a$$

$$= \int_a^b \frac{x - x_1}{x_0 - x_1}\,dx \qquad x_1 = b$$

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$$= \int_{a}^{b} \frac{x - a}{b - a} \, dx$$

$$= \frac{1}{b-a} \int (x - a) \, dx$$

$$= \frac{1}{b-a} \left[ \frac{x^2}{2} - ax \right]_a^b$$

$$= \frac{1}{b-a} \left[ \frac{b^2}{2} - ab - \frac{a^2}{2} + a^2 \right]$$

$$= \frac{1}{b-a} \left[ \frac{b^2 - 2ab - a^2 + 2a^2}{2} \right]$$

$$= \frac{1}{b-a} \left[ \frac{-(a^2 - 2ab + b^2)}{2} \right]$$

$$= \frac{1}{b-a} \left[ \frac{-(a-b)^2}{2} \right]$$

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$$= \frac{b-a}{2}$$

$$\sigma_{\perp} = \int_a^b f_{\perp}(x)\, dx$$

$$= \int_a^b \frac{x - x_0}{x_{\perp} - x_0}\, dx$$

$$= \int_a^b \frac{x-a}{b-a}\, dx$$

$$= \frac{1}{b-a} \int_a^b (x-a)\, dx$$

$$= \frac{1}{b-a} \left[ \frac{x^2}{2} - ax \right]_a^b$$

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$$= \frac{1}{z-a}\left[\frac{z^2}{2} - az - \frac{a^2}{2} + a^2\right]$$

$$= \frac{1}{z-a}\left[\frac{z^2 - 2az - a^2 + 2a^2}{2}\right]$$

$$= \frac{1}{z-a}\cdot\frac{(a-z)^2}{2}$$

$$= \frac{z-a}{2}$$

$$\therefore I_1(z) = \sigma_0 z(x_0) + \sigma_1 z(x_1)$$

$$= \frac{z-a}{2} z(a) + \frac{z-a}{2} z(z)$$

$$= \frac{z-a}{2}\left[z(a) + z(z)\right]$$

[Proved]

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# Simpson's rule :

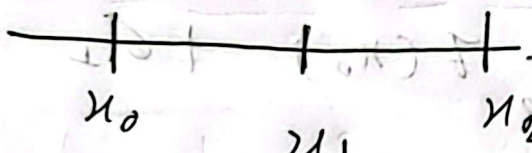
$$I_2(f) = \int_a^b P_2(x)\,dx$$

$$= \int_a^b [l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2)]\,dx$$

$$= C_0 f(x_0) + C_1 f(x_1) + C_2 f(x_2)$$

Since $n = 3$, node $= 2 + 1$

$= 3$




$$x_1 = x_0 + h$$

$$m = x_1 = \frac{a+b}{2}$$

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$$G_0 = \int_a^b \lambda_0(x)\, dx$$

$$= \int_a^b \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}\, dx$$

$$= \int_a^b \frac{(x-m)(x-\tau)}{(a-m)(a-\tau)}\, dx$$

$$= \frac{1}{(a-m)(a-\tau)} \int_a^b (x-m)(x-\tau)\, dx$$

$$= \frac{1}{(a-m)(a-\tau)} \int_a^b (x^2 - mx - \tau x + \tau m)\, dx$$

$$= \frac{1}{(a-m)(a-\tau)} \left[ \frac{x^3}{3} - \frac{mx^2}{2} - \frac{\tau x^2}{2} + \tau m x \right]_a^b$$

$$= \frac{1}{(a-m)(a-\tau)} \left[ \frac{b^3}{3} - \frac{b^2 m}{2} - \frac{b^2 \tau}{2} + b\tau m - \frac{a^3}{3} + \frac{a^2 m}{2} + \frac{a^2 \tau}{2} - a\tau m \right]$$

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$$= \frac{1}{\left(a-\frac{a+t}{2}\right)(a-t)}\left[\frac{t^3}{3} - \frac{t^2\times\left(\frac{a+t}{2}\right)}{2} - \frac{t^3}{2}\right.$$

$$\left. + t^2\times\frac{(a+t)}{2} - \frac{a^3}{3} + \frac{a^2\left(\frac{a+t}{2}\right)}{2} + \frac{a^2t}{2} - at\times\frac{(a+t)}{2}\right]$$

$$= \frac{1}{\left(\frac{a-t}{2}\right)(a-t)}\left[\frac{t^3}{3} - \frac{at^2}{4} - \frac{t^3}{4} - \frac{t^3}{2} + \frac{at^2}{2}\right.$$

$$\left. + \frac{t^3}{2} - \frac{a^3}{3} + \frac{a^3}{4} + \frac{a^2t}{4} + \frac{a^2t}{2} - \frac{a^2t}{2} - \frac{at^2}{2}\right]$$

$$= \frac{2}{(a-t)^2}\times\left[\frac{t^3}{12} - \frac{at^2}{4} - \frac{a^3}{12} + \frac{a^2t}{4}\right]$$

$$= \frac{2}{(a-t)}\left[\frac{t^3 - 3at^2 - a^3 + 3a^2t}{12}\right]$$

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$$= \frac{2}{(a-z)} \left[ \frac{(z-a)^3}{12} \right]$$

$$= \frac{1}{(a-z)} \times \frac{-(a-z)^2}{6}$$

$$= \frac{z-a}{6}$$

$$G_1 = \int_a^z \lambda_1(x)\, dx$$

$$= \int_a^z \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}\, dx$$

$$= \int_a^a \frac{(x-a)(x-z)}{(m-a)(m-z)}\, dx_1$$

$$= \frac{1}{(m-a)(m-z)} \int (x-a)(x-z)\, dx$$

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$$= \frac{1}{(m-a)(m-z)} \int_a^z (x^2 - ax - zx + az)\, dx$$

$$= \frac{1}{(m-a)(m-z)} \left[ \frac{x^3}{3} - \frac{ax^2}{2} - \frac{zx^2}{2} + azx \right]_a^z$$

$$= \frac{1}{(m-a)(m-z)} \left[ \frac{z^3}{3} - \frac{az^2}{2} - \frac{z^3}{2} + az^2 - \frac{a^3}{3} + \frac{a^3}{2} + \frac{a^2 z}{2} - a^2 z \right]$$

$$= \frac{1}{\left(\frac{a+z}{2} - a\right)\left(\frac{a-z}{2} - z\right)} \left[ -\frac{z^3}{6} + \frac{az^2}{2} + \frac{a^3}{6} + \frac{a^2 z}{2} \right]$$

$$= \frac{1}{\frac{(a+z-2a)}{2} \times \frac{(a+z-2z)}{2}} \left[ \frac{-z^3 + 3az^2 + a^3 - 3a^2 z}{6} \right]$$

$$= \frac{4}{(z-a)(a-z)} \times \frac{1}{6} (a-z)^3$$

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$$= \frac{2}{3} \times (2-a)$$

$$6_2 = \int_a^2 l_2(x)\, dx$$

$$= \int_a^2 \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}\, dx$$

$$= \int_a^2 \frac{(x-a)(x-m)}{(2-a)(2-m)}\, dx$$

$$= \frac{1}{(2-a)(2-m)} \int_a^2 (x-a)(x-m)\, dx$$

$$= \frac{1}{(2-a)(2-m)} \int_a^2 (x^2 - ax - mx + am)\, dx$$

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$$= \frac{1}{(z-a)(b-z)} \left[ \frac{x^3}{3} - \frac{ax^2}{2} - \frac{mx^2}{2} + cmx \right]_a^z$$

$$= \frac{1}{(z-a)\left(z-\frac{a+z}{2}\right)} \left[ \frac{z^3}{3} - \frac{az^2}{2} - \frac{z^2}{2} \times \frac{(a+z)}{2} \right.$$

$$+ cmx \frac{(a+z)}{2} - \frac{a^3}{3} + \frac{a^3}{2} + \frac{a^2}{2} \times \frac{(a+z)}{2}$$

$$\left. - a^2 \times \frac{(a+z)}{2} \right]$$

$$= \frac{1}{(z-a)\left(\frac{2z-a-z}{2}\right)} \left[ \frac{z^3}{3} - \frac{az^2}{2} - \frac{az^2}{4} - \frac{z^3}{4} + \frac{a^2 z}{2} \right.$$

$$\left. + \frac{az^2}{2} - \frac{a^3}{3} + \frac{a^3}{2} + \frac{a^3}{4} + \frac{a^2 z}{4} - \frac{a^3}{2} - \frac{a^2 z}{2} \right]$$

$$= \frac{2}{(z-a)^2} \left[ \frac{z^3 - az^2 + a^2 z - a^3}{12} \right]$$

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$$= \frac{1}{(z-a)^2} \times \frac{(z-a)^3}{6}$$

$$= \frac{1}{6}(z-a)$$

$$I_2(z) = 6_0 z(x_0) + 6_2 z(x_1) + 6_2 z(x_2)$$

$$= 6_0 z(a) + 6_2 z(m) + 6_2 z(z)$$

$$= \frac{z-a}{6} z(a) + \frac{2}{3}(z-a) z(m) + \frac{z-a}{6} z(z)$$

$$= \frac{z-a}{6}\left[z(a) + 4z\left(\frac{a+z}{2}\right) + z(z)\right]$$

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